Artificial Intelligence and Orthopaedics: An Introduction for Clinicians.

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CLINICAL & AGENTIC AI TAKEAWAY

AI in orthopaedics can improve care and reduce physician burnout, but ethical issues and clinical superiority need to be addressed.

ABSTRACT

➤ Artificial intelligence (AI) provides machines with the ability to perform tasks using algorithms governed by pattern recognition and self-correction on large amounts of data to narrow options in order to avoid errors. ➤ The 4 things necessary for AI in medicine include big data sets, powerful computers, cloud computing, and open source algorithmic development. ➤ The use of AI in health care continues to expand, and its impact on orthopaedic surgery can already be found in diverse areas such as image recognition, risk prediction, patient-specific payment models, and clinical decision-making. ➤ Just as the business of medicine was once considered outside the domain of the orthopaedic surgeon, emerging technologies such as AI warrant ownership, leverage, and application by the orthopaedic surgeon to improve the care that we provide to the patients we serve. ➤ AI could provide solutions to factors contributing to physician burnout and medical mistakes. However, challenges regarding the ethical deployment, regulation, and the clinical superiority of AI over traditional statistics and decision-making remain to be resolved.

Study Overview & Background

➤ Artificial intelligence (AI) provides machines with the ability to perform tasks using algorithms governed by pattern recognition and self-correction on large amounts of data to narrow options in order to avoid errors. ➤ The 4 things necessary for AI in medicine include big data sets, powerful computers, cloud computing, and open source algorithmic development. ➤ The use of AI in health care continues to expand, and its impact on orthopaedic surgery can already be found in diverse areas such as image recognition, risk prediction, patient-specific payment models, and clinical decision-making. ➤ Just as the business of medicine was once considered outside the domain of the orthopaedic surgeon, emerging technologies such as AI warrant ownership, leverage, and application by the orthopaedic surgeon to improve the care that we provide to the patients we serve. ➤ AI could provide solutions to factors contributing to physician burnout and medical mistakes. However, challenges regarding the ethical deployment, regulation, and the clinical superiority of AI over traditional statistics and decision-making remain to be resolved.

Methodology & Clinical Focus

Study Classification: other. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

AI in orthopaedics can improve care and reduce physician burnout, but ethical issues and clinical superiority need to be addressed.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

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